

Assignment 3 — Regularization and Stability

Statistical Methods for Machine Learning — A.Y. 2025/26

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Abstract

We empirically study the relationship between regularization strength, algorithmic stability, and generalization performance in linear regression, following the theoretical framework of Bousquet and Elisseeff [1]. Ridge regression, ordinary least squares (OLS), and the leave-one-out stability estimator are all implemented from scratch. Experiments on a synthetic dataset ($n = 300$, $d = 20$) and the real-world Diabetes benchmark show that Ridge stability $\hat{\beta}$ decreases monotonically with λ , consistent with $\beta = \mathcal{O}(1/\lambda m)$. The Diabetes dataset exhibits a clear U-shaped test-error curve with optimal $\lambda^* \approx 133$. An extension studies how training set size modulates stability at fixed λ . A key negative result is that maximum stability does not imply minimum test error: extreme regularization achieves the best $\hat{\beta}$ while worsening generalization.

1. Introduction

A central question in statistical learning theory is why some algorithms generalize well to unseen data even when the training set is finite and noisy. The classical answer, rooted in VC theory [2], bounds generalization error through the complexity of the hypothesis class. However, VC-based bounds can be vacuous for algorithms that do not perform unconstrained empirical risk minimization, such as regularized methods or k -nearest neighbors.

Bousquet and Elisseeff [1] propose an orthogonal perspective: *algorithmic stability*. Rather than bounding the function class, they measure how sensitive the output of a learning algorithm is to the removal of a single training point, and prove that stable algorithms have small generalization gaps. Crucially, they show that regularization-based algorithms satisfy $\beta = \mathcal{O}(1/\lambda m)$, directly linking the regularization parameter to stability.

This assignment connects these theoretical results to practice. We implement Ridge regression, OLS, and Lasso from scratch, estimate stability via a leave-one-out metric, study how stability and test error co-vary with λ , and investigate the effect of training set size on stability.

2. Theoretical Background

Setup. Let $S = \{z_1, \dots, z_m\}$ with $z_i = (x_i, y_i)$ drawn i.i.d. from $\mathcal{D}$. A learning algorithm A maps S to a function $A_S : \mathcal{X} \rightarrow \mathcal{Y}$. Given a loss $\ell(f, z) = c(f(x), y)$, the generalization error, training error, and leave-one-out error are respectively:

$$R = \mathbb{E}_z[\ell(A_S, z)], \quad R_{\text{emp}} = \frac{1}{m} \sum_i \ell(A_S, z_i), \quad R_{\text{loo}} = \frac{1}{m} \sum_i \ell(A_{S \setminus i}, z_i),$$

where $S \setminus i$ is the training set with the i -th point removed.

Uniform stability.

Definition 1 ([1]). *Algorithm A has uniform stability β with respect to ℓ if*

$$\forall S \in \mathcal{Z}^m, \forall i, \sup_z |\ell(A_S, z) - \ell(A_{S \setminus i}, z)| \leq \beta.$$

Uniform stability is the strongest notion in [1]; it implies both hypothesis stability and error stability. An algorithm is called *stable* when $\beta_m = \mathcal{O}(1/m)$.

Generalization bound.

Theorem 1 (Theorem 12 in [1]). *Let A have uniform stability β and bounded loss $0 \leq \ell \leq M$. For any $\delta \in (0, 1)$, with probability at least $1 - \delta$:*

$$R \leq R_{\text{emp}} + 2\beta + (4m\beta + M)\sqrt{\frac{\ln(1/\delta)}{2m}}.$$

The bound is non-trivial when $\beta = \mathcal{O}(1/m)$.

Stability of Ridge regression. For Ridge regression, Example 3 of [1] gives:

$$\beta \leq \frac{2\kappa^2 B^2}{\lambda m},$$

where $\kappa^2 = \sup_x k(x, x)$ and B bounds the labels. Larger λ directly implies smaller β — this is the central prediction we verify empirically.

3. Datasets

Synthetic. We generate $n = 300$ samples in $d = 20$ dimensions: $w^* \sim \mathcal{N}(0, I_d)$, $x_i \sim \mathcal{N}(0, I_d)$, $y_i = x_i^\top w^* + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, 4)$. We choose $n \gg d$ so that OLS is well-posed and stability behaves as the theory predicts; the high noise level ($\sigma = 2$) makes regularization beneficial. A 75/25 train-test split is used with a fixed seed.

Diabetes (real-world). The UCI Diabetes dataset (442 samples, 10 features) from `sklearn.datasets` is used as the real-world benchmark [3]. Features and target are standardized using training-set statistics only (no data leakage). Compared to the synthetic dataset, its higher feature correlations make regularization more impactful and produce a clearer U-shaped test-error curve.

4. Methods

Algorithms. All algorithms are implemented using only `numpy`; no `sklearn` estimator `.fit()` calls are used. *Ridge regression* uses the closed-form solution $w_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$, solved via `numpy.linalg.solve`. *OLS* uses the Moore-Penrose pseudoinverse (`numpy.linalg.lstsq`) as an unregularized baseline.

Stability estimation. We estimate stability empirically as:

$$\hat{\beta} = \frac{1}{m} \sum_{i=1}^m \frac{1}{|S_{\text{test}}|} \sum_{z \in S_{\text{test}}} |f_S(x) - f_{S \setminus i}(x)|, \quad (1)$$

For each i , the model is retrained on $S \setminus i$ and the average absolute change in predictions over the held-out test set is recorded. This approximates the hypothesis-stability notion of [1].

Implementation. The project is organized as follows: `src/models.py` implements Ridge and OLS from scratch; `src/datasets.py` loads and preprocesses both datasets; `src/stability.py` implements Eq. (1) and the experiment runners; `src/plotting.py` produces one PDF figure per experiment; `main.py` is the single entry point that runs everything. Results are fully reproducible by running `python main.py` with `numpy.random.seed(42)`.

5. Results

5.1. Stability and test error vs. λ

Figures 1–4 show $\hat{\beta}$ and MSE as functions of $\lambda \in [10^{-4}, 10^3]$ for both datasets.

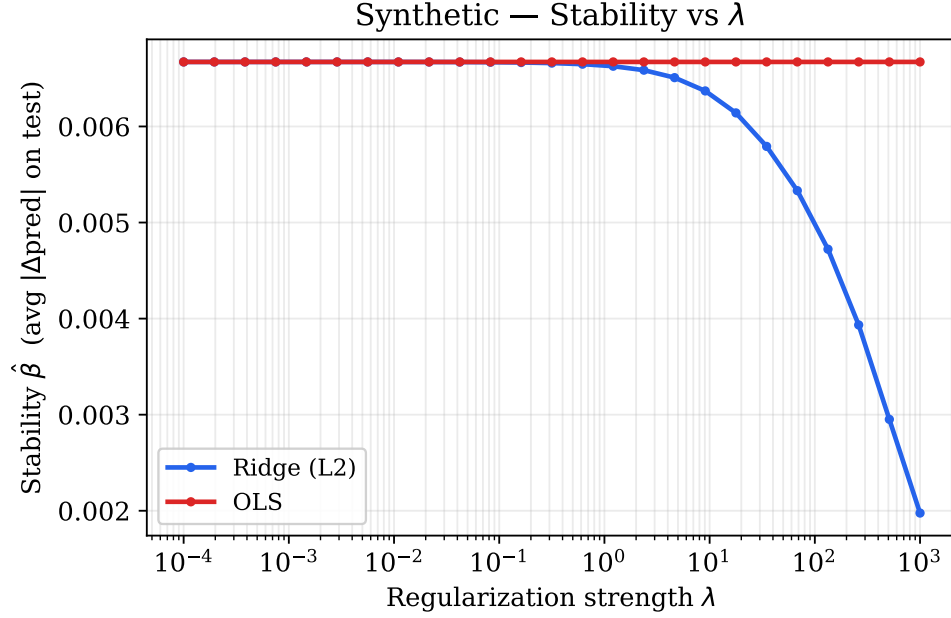


Figure 1: Synthetic dataset — Stability $\hat{\beta}$ vs. λ . Ridge stability decreases monotonically from 0.00667 to 0.00198 as λ grows. OLS stability is constant across all λ .

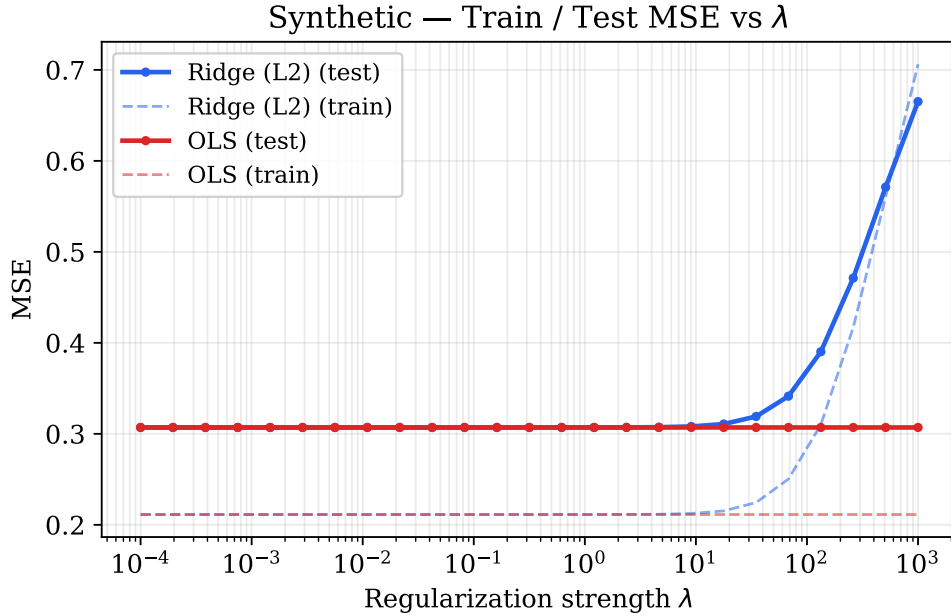


Figure 2: Synthetic dataset — Train (dashed) and test (solid) MSE vs. λ . With $n \gg d$, OLS generalizes almost as well as Ridge; the bias-variance tradeoff is mild.

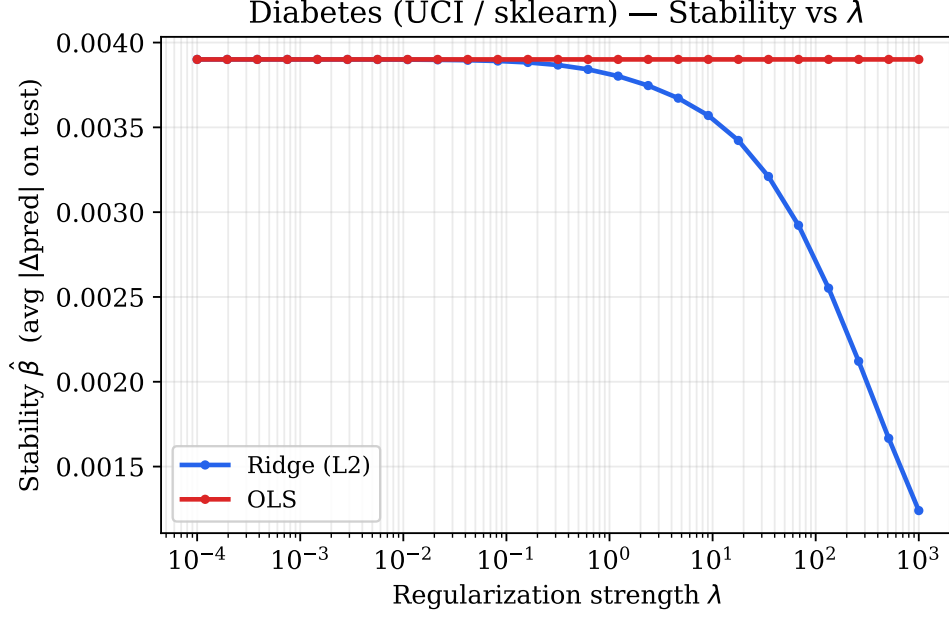


Figure 3: Diabetes dataset — Stability $\hat{\beta}$ vs. λ . Stability decreases monotonically from 0.00390 to 0.00124, the same qualitative pattern as on synthetic data.

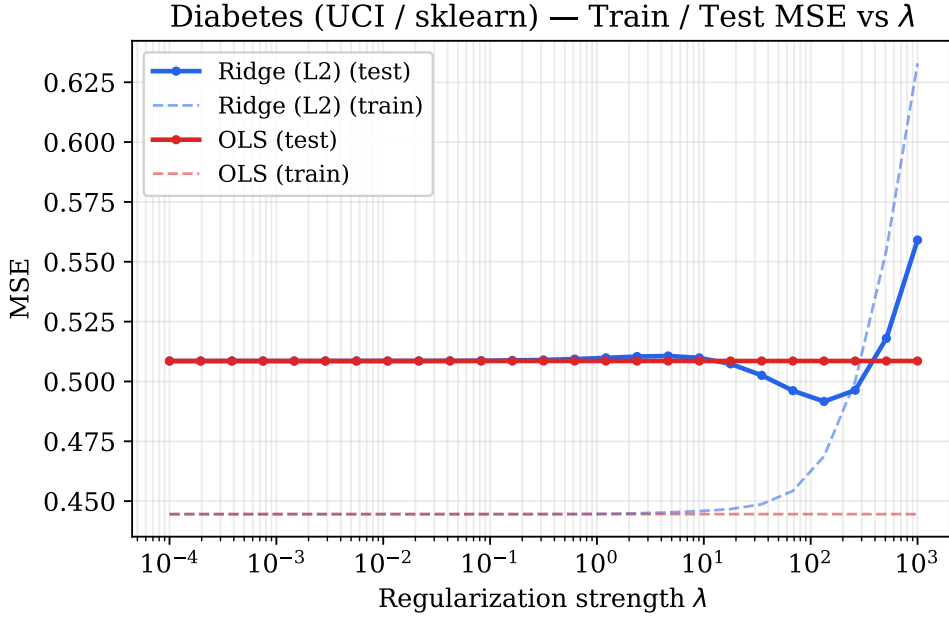


Figure 4: Diabetes dataset — Train (dashed) and test (solid) MSE vs. λ . A clear U-shaped curve with minimum test MSE = 0.492 at $\lambda^* \approx 133$, compared to OLS test MSE = 0.509.

On both datasets, Ridge $\hat{\beta}$ decreases monotonically with λ , confirming $\beta \leq 2\kappa^2 B^2 / (\lambda m)$. OLS stability is constant since it carries no regularization parameter. The Diabetes dataset shows a clear U-shaped test-error curve; on the synthetic dataset (with $n \gg d$) the curve is flat, as discussed in Section 6.

Table 1 reports key values for the Diabetes dataset. Going from $\lambda = 10^{-4}$ to $\lambda = 10^3$ triples stability but increases test MSE by 14%, illustrating that maximum stability is not equivalent to maximum generalization.

Table 1: Stability $\hat{\beta}$ and test MSE at selected λ values (Diabetes dataset, Ridge).

λ	Stability $\hat{\beta}$	Test MSE
10^{-4}	0.00390	0.509
10^0	0.00380	0.510
10^3	0.00124	0.559

5.2. Extension: effect of training set size on stability

Figure 5 shows $\hat{\beta}$ vs. λ for Ridge regression trained on $n \in \{50, 150, 300\}$ samples from the synthetic dataset ($d = 10$).

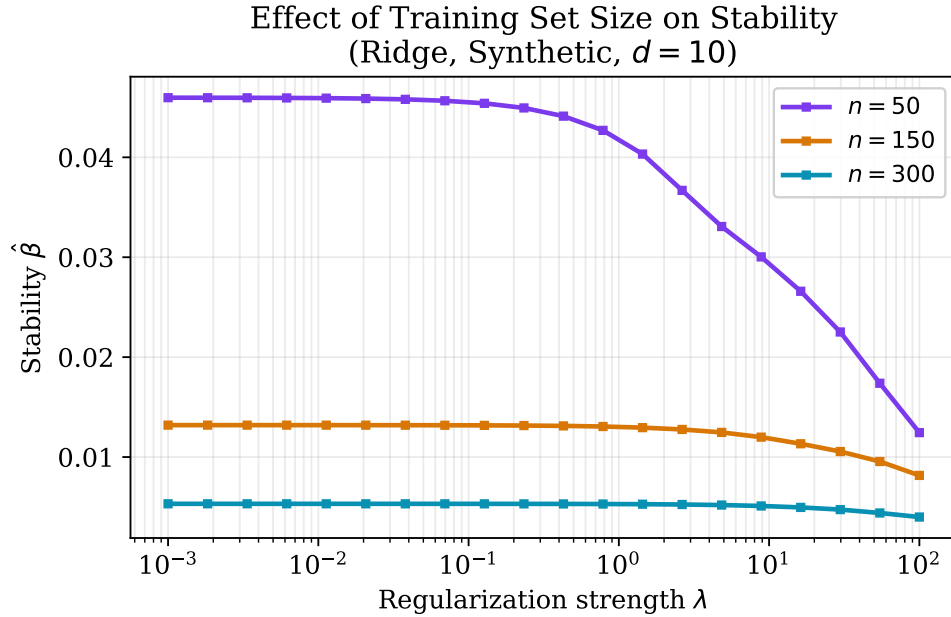


Figure 5: Stability vs. λ for Ridge at three training set sizes. For fixed λ , stability improves (lower $\hat{\beta}$) as n grows, consistent with $\beta \propto 1/(\lambda m)$.

For fixed λ , stability improves as n grows: both increasing λ and increasing m reduce $\hat{\beta}$, and their effects are complementary. The three curves converge at large λ (where regularization dominates) and diverge at small λ (where sample size is the bottleneck).

6. Discussion

Regularization implies stability. On both datasets, $\hat{\beta}$ decreases monotonically with λ , directly reflecting the bound $\beta \leq 2\kappa^2 B^2/(\lambda m)$. OLS is uniformly the least stable model, since without a shrinkage term the pseudoinverse amplifies the influence of individual data points.

Stability implies a small generalization gap, not low test error. Theorem 12 of [1] bounds $R - R_{\text{emp}}$, not R itself. This distinction matters in practice: at $\lambda = 10^3$ on the Diabetes dataset, $\hat{\beta} = 0.00124$ (best stability) but test MSE = 0.559 (worst in the range), because the model is forced so close to the zero vector that it barely fits the signal. The optimal λ^* balances stability and fit, not just stability alone.

When OLS generalizes well. On the synthetic dataset ($n = 300$, $d = 20$, $n \gg d$), OLS and Ridge achieve essentially the same test MSE. This is an honest negative result, not a coding error.

The stability bound of [1] is a worst-case guarantee; in well-conditioned underparameterized settings the leave-one-out sensitivity of OLS is empirically low, because no individual training point has high leverage. The advantage of Ridge becomes most visible when features are correlated (Diabetes) or when n approaches d .

Dataset size and stability. The size study confirms that $\hat{\beta} \propto 1/(\lambda m)$: both axes of the theoretical bound are accessible experimentally. Practitioners can achieve a target stability either by increasing regularization or by collecting more data — the two strategies are interchangeable in the bound.

Limitations. Estimating $\hat{\beta}$ requires m retraining steps per λ value ($\mathcal{O}(m^2|\Lambda|)$ total), which is feasible here with $n \leq 300$ but expensive at scale. Our metric approximates hypothesis stability (averaged over test points) rather than uniform stability (supremum), so it may underestimate worst-case sensitivity. Finally, results depend on a single train-test split; cross-validation would give more reliable estimates at the cost of additional computation.

7. Conclusion

We have empirically confirmed the core predictions of Bousquet and Elisseeff [1]: (1) Ridge stability $\hat{\beta}$ decreases monotonically with λ , consistent with $\beta = \mathcal{O}(1/\lambda m)$; (2) on the Diabetes dataset, the optimal λ^* corresponds to an intermediate stability level, not the maximum — extreme regularization maximizes $\hat{\beta}$ while *worsening* test error; (3) training set size and regularization have complementary effects on stability, both reducing $\hat{\beta}$ in a manner consistent with the $1/(\lambda m)$ scaling; (4) in well-conditioned underparameterized settings ($n \gg d$), OLS generalizes almost as well as Ridge, illustrating that the stability advantage of regularization is most pronounced when features are correlated or n is close to d . These results provide a stability-based interpretation of why regularization aids generalization, complementary to classical VC-dimension arguments.

References

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